# Optimizing Canada's Conservation Strategy: OECM Candidate Identification in Átl'ka7tsem / Howe Sound Biosphere Region

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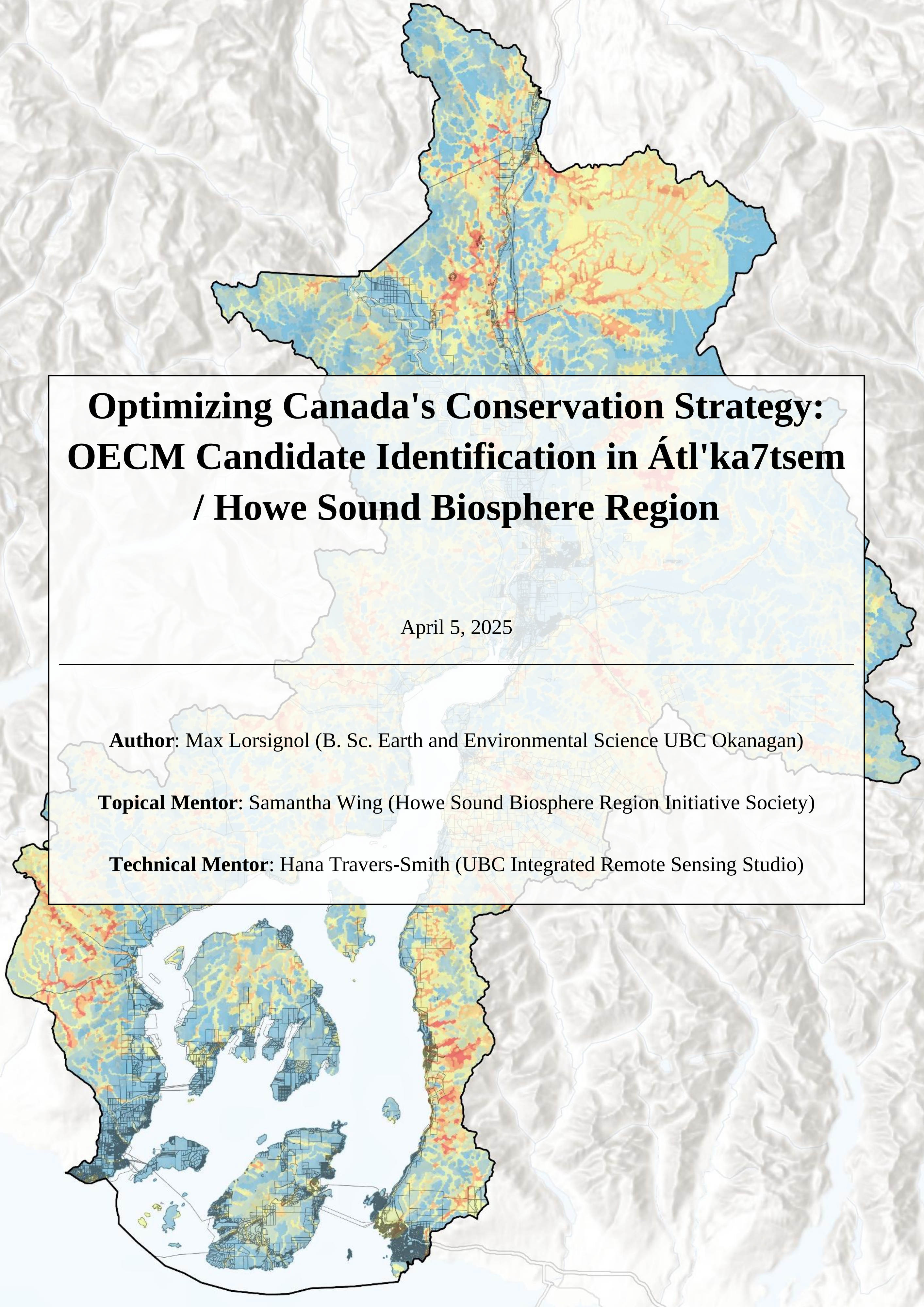

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## Abstract

To support Canada's ambitious conservation commitments of protecting 25% of its terrestrial and marine ecosystems by 2025, and 30% by 2030, new and flexible conservation approaches are necessary. Other Effective Area-Based Conservation Measures (OECMs) represent an opportunity to achieve these targets by recognizing biodiversity conservation occurring indirectly through existing land uses rather than explicitly through protected area designation. This research developed a systematic methodology to identify candidate OECM parcels in Átl'k̲a7tsem/Howe Sound Biosphere Region, British Columbia. By integrating ecological variables with governance and zoning data into a weighted suitability model, the study demonstrated a scalable framework for OECM candidate parcel selection. Results highlighted 260 candidate parcels across three regional districts, emphasizing parcels whose governance structures align well with long-term conservation goals. The developed methodology addresses critical limitations of traditional qualitative approaches, enhancing conservation planning efficiency and scalability. Ultimately, this study provides a replicable model to facilitate Canada's OECM identification process, promoting biodiversity conservation and contributing meaningfully to Canada's climate and ecological objectives.

**Keywords**: Other Effective area-based Conservation Measures OECM, Protected Areas, BC Parcel Fabric, Weighted Suitability Model, 30-by-30, Átl'k̲a7tsem/Howe Sound Biosphere Region

## 1. Introduction

As ecosystems continue to face mounting pressures from habitat loss, species endangerment, and the decline of ecosystem services, governing bodies, institutes, non-profits and other organizations are taking action. In order to combat these pressures, the Convention on Biological Diversity (CBD) has initiated a strategic plan to conserve at least 17% of terrestrial and inland water, and 10% of coastal and marine areas (MacKinnon et al., 2015). Specifically, the Canadian government has created its own goal: to recognize 25% of Canadian marine and terrestrial area as Protected Areas and OECMs by the year 2025 with an even more ambitious vision of 30% by the year 2030 (Trudeau, 2021), recognizing that an 'all in' effort is necessary to achieve such a goal (Improvising BC's Approach to Claiming Other Conserved Areas, n.d.). In order to achieve this goal, the concept of Other Effective area-based Conservation Methods (OECM) was introduced. OECMs differ from Protected Areas (PA) by changing the primary purpose from biodiversity conservation to more general, long-term conservation as a byproduct of changes in land use management (IUCN WCPA Task Force on OECMs, 2019). A shift in the primary purpose of land conservation allows for more a diverse range of lands to qualify as a recognized conserved land, improving the chances of reaching climate goals.

## *1.1. Defining OECM*

The International Union for Conservation of Nature (IUCN) defines an OECM as 'an area that is geographically defined distinct from Protected Areas (PA) but managed in a way that yield positive, sustained and long-term outcomes for biodiversity conservation' (P. J. Fitzsimons & Stolton, n.d.). The Government of Canada further elaborates on this definition on OECMs by adding "support the long-term outcomes for the in-situ conservation of biodiversity, with associated ecosystem functions and services and where applicable, cultural, spiritual, socio-economical and other locally relevant values" (PDF, n.d.) The difference between a Protected Area and an OECM is that the primary focus is not of conservation, but rather that an effective and long-lasting in-situ conservation of biodiversity is delivered, even if biodiversity is not the primary conservation objective (P. J. Fitzsimons & Stolton, n.d.). In-situ conservation refers to the conservation of ecosystems and natural habitats of species in their natural habitat, in the surroundings where they have developed their distinctive properties (Maxted, 2001).

## *1.2. OECM Process*

The implementation and recognition of OECMs in Canada and globally is a crucial mechanism to meet international biodiversity targets (Gurney et al., 2021), particularly those outlined in the CBD Kunming-Montreal Global Biodiversity Framework (CBD, 2022). The process of identifying such areas is largely guided by qualitative evaluation of biodiversity conservation effectiveness, land-use designations, governing mechanisms and the capacity to maintain a long-term protection. Federal guidance from the Canadian Council on Ecological Areas (CCEA) emphasize the need to properly assess these sites against the broad criteria provided by the Decision Support Tool (2020-2030 Canadian Council on Ecological Areas Strategic Framework.Pdf, n.d.; MacKinnon et al., 2015). This criterion specifically includes: defined geographical space, sustained biodiversity conservation, effective governance and monitoring capacity (CCEA, 2019; MacKinnon et al. 2015).

Current methods for identifying OECMs in Canada rely heavily on qualitative assessments and stakeholder consultations. Key criteria include the area's contribution to biodiversity conservation, land-use designation, governance, and its capacity to deliver long-term conservation outcomes (Environment and Climate Change Canada, n.d.). However, these methods face limitations; a lack of comprehensive, region-specific biodiversity data weaken the case for recognizing eligible OECMs; existing tools often struggle to represent complex land use patterns and ecosystems in a way that fully captures conservation potential; and differing economic, cultural, or environmental priorities among stakeholders can slow the process of reaching a consensus on OECM recognition (Lemieux et al., 2022).

In response to these challenges, this proposal offers an enhanced approach to OECM identification, using Átl'ḵa7tsem/Howe Sound Biosphere Region as a case study. By incorporating a range of Geographic Information System (GIS) and data analytical tools, the objective is to create a more replicable and systematic process for evaluating potential OECMs. This approach will use the CBD's decision support tool in conjunction with the Canadian Government DST, supplemented by mapped biodiversity and ecosystem data, to pinpoint areas that are more likely to be eligible for OECM recognition, and deliver substantial conservation benefits without requiring biodiversity conservation to be the primary objective of the land. Ultimately, this research will aim to create detailed maps that define geographic boundaries and land use designations for OECMs in Howe Sound by focusing on areas where biodiversity conservation is an indirect outcome of other activities. This method offers a pragmatic way to contribute to Canada's conservation goals, while acknowledging the complexity of land use in diverse, multi-purpose landscapes.

## 2. Site Summary and Data Description

Átl'ḵa7tsem/Howe Sound, a typical fjord, is an inlet of the Salish Sea and lies between British Columbia's Lower Mainland and the Sunshine Coast. The region's landscape is marked by a variety of natural features including mountains, forests, estuaries, alpine lakes, dormant volcanic formations, and rugged coastlines (Geography of British Columbia | The Canadian Encyclopedia, n.d.). Extending approximately 42 kilometers from West Vancouver to Squamish, Howe Sound contains numerous islands, such as Bowen, Keats, Anvil, and Gambier Islands (Mackie, n.d.). The fjord supports many known marine and terrestrial species and thrives within diverse ecosystems that provide essential ecological services and contribute to its high conservation, recreational, and aesthetic value, and is enjoyed by residents and visitors alike. (OceanWatch - Howe Sound Report 2020 - Online, n.d.).

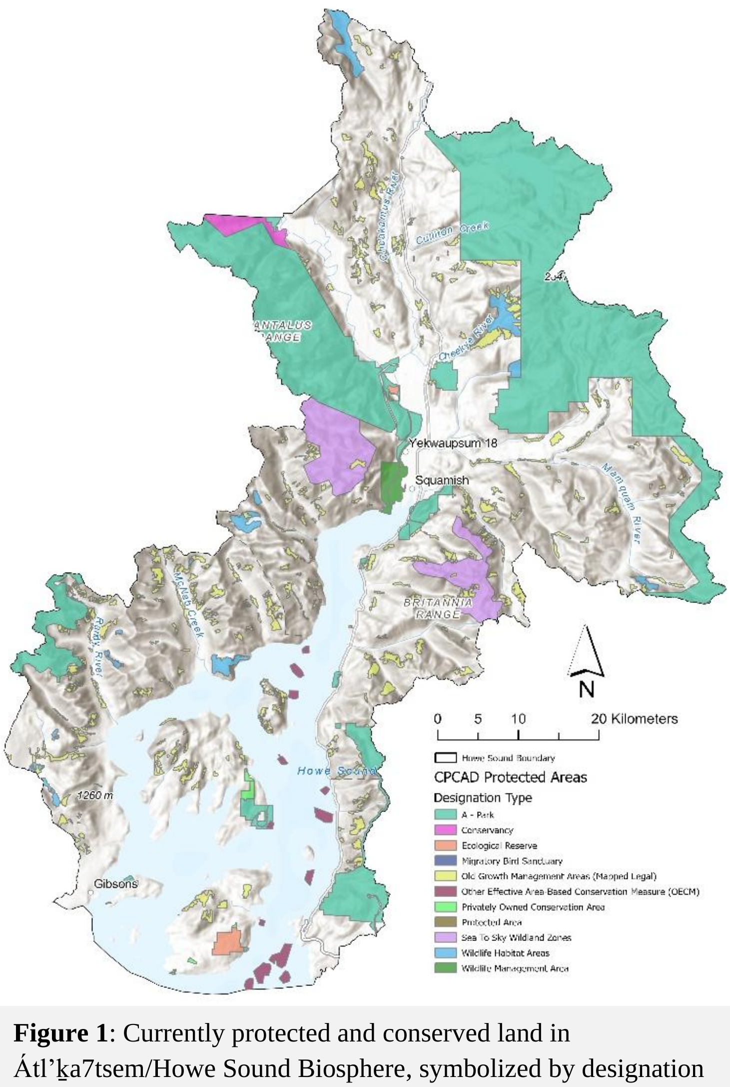


**Figure 1**: Currently protected and conserved land in Átl'ḵa7tsem/Howe Sound Biosphere, symbolized by designation type, based on the CPCAD dataset.

The Átl'ḵa7tsem/Howe Sound UNESCO Biosphere Region covers 218,723 hectares (2,187 square kilometers), with terrestrial environments accounting for 84% and marine environments comprising the remaining 16% (Átl'ḵa7tsem / Howe Sound, n.d.). A significant portion, roughly 40% of the terrestrial area, is managed for conservation purposes such as Tantalus Park, Garibaldi Park and the Sea to Sky Wildlands (Mackie, n.d.). The spatial distribution of these conserved lands is illustrated in Figure 1. The remaining land is used for mixed activities including residential development and sustainable resource use. The biosphere is primarily within the unceded territory of the Sḵwx̱wú7mesh Úxwumixw (Squamish Nation) and other Indigenous communities. Management of the region involves a collaborative stewardship between Indigenous peoples and the Province of British Columbia. With a population of approximately 50,000, the area includes several vibrant communities such as Squamish, Gibsons, and Horseshoe Bay, along with smaller settlements and seasonally inhabited islands. Historically impacted by unsustainable resource extraction, Howe Sound has seen a shift toward conservation-focused initiatives since the mid-1970s, with sustainable forestry and ecotourism now playing key roles in the local economy (Mackie, n.d.).

Átl’ḵa7tsem/Howe Sound offers an ideal case study for identifying potential new OECMs due to its diverse ecosystems, unique conservation history, and collaborative management approach (Mackie, n.d.). Its combination of terrestrial, marine, and freshwater ecosystems, coupled with strong community involvement and Indigenous stewardship, makes it a prime candidate for further conservation designations. By building on the existing protected areas and incorporating new OECM strategies, Howe Sound can contribute significantly to Canada’s ambitious 30 by 30 goals, offering valuable insights into balancing biodiversity protection with sustainable economic activities.

## *2.1. Datasets*

To effectively identify and prioritize potential OECMs in the Howe Sound Biosphere Region, a variety of spatial datasets were utilized. At the foundation is the *AHSBR Biosphere Boundary* which delineates the geographic scope of the analysis. This boundary serves as the reference for all the subsequent data as well as the layer used to clip the data.

Ecological and biodiversity information, a key aspect to OECM recognition, is captured through several datasets. The *Critical Habitat for Federally-listed Species at Risk* and the *Species and Ecosystems at Risk* layers provide vital information and insight into habitat distribution and the potential presence of vulnerable species. These are critical in pinpointing areas where conservation efforts could have the greatest biodiversity impact. In addition to this data, the *Vegetation Resource Inventory (VRI)* is used to gain insight on the forest composition, structure and forest age, essential for assessing habitat quality and ecosystem connectivity and integrity.

Physical and hydrological features are also considered in this analysis. The *Digital Elevation Model (DEM)* is used to delineate steeply sloped areas in the Sound which is useful for development suitability and habitat. The *Freshwater Stream Network* and *Wetlands* layers capture the aquatic ecosystems which are vital for hydrological connectivity and biodiversity. Additionally, the *Community Watersheds* were included to highlight areas that contained social significance due to their its role in providing clean water to human populations and support to ecosystem health.

**Table 1. List of the data sets used in this analysis, including the title of the dataset, a brief description and the source. The majority of the layers are sourced from the BC Data Catalog.**

| Data Title | Description | Source |
|---|---|---|
| AHSBR Biosphere Boundary | Boundary of the Átl'ka7tsemHowe Sound UNESCO Biosphere Region based on the Howe Sound Cumulative Effects boundary. | HSBRIS |
| Critical Habitat for Federally-Listed Species at Risk | Contains habitat polygons for species at risk in Canada. | Environment Canada |
| Species and Ecosystems at Risk (CDC) | Known locations of species and ecological communities at risk & masked layer that generalizes the locations of those species and communities at risk. | Ecosystems |
| ParcelMap BC Parcel Fabric | A subset of the complete ParcelMap BC data, consisting of the parcel fabric and attributes clipped to the extent of Howe Sound. | Data Systems and Services (DataBC) |
| Canadian Protected and Conserved Areas Database (CPCAD) | Contains spatial and attribute data on marine and terrestrial protected areas including current OECM. | ECCC |
| Vegetation Resource Inventory (VRI) | A geospatial forest inventory dataset. Attributes include age, species, volume, and height. | Forest Analysis and Inventory Branch |
| Digital Elevation Model (DEM) | Terrain Resource Information Management (TRIM) data and converted into the Canadian Digital Elevation Data (CDED). | GeoBC Branch |
| Freshwater Stream Network | Represents fresh water streams and rivers. | GeoBC Branch |
| Wetlands | All the wetland polygons of the province of BC | GeoBC Branch |
| Community Watersheds | Boundaries of areas designated as community watersheds by the government. | Aquatic Ecosystems |
| Zoning Layers | Range of zoning layers from a variety of sources. | Various[1] |

[1]Zoning layers source from West Vancouver, Gibsons, Sunshine Coast, Island Trust, Squamish and Lions Bay

Finally, land-use and governance context are derived from the *BC Parcel Fabric* and *Zoning* layers. These datasets enable the spatial identification of the final selection of OECM candidate parcels, and aid in the understanding of the parcel attributes like ownership, tenure and permissible land-uses. Understanding the underlaying governance of the parcel undergoing OECM consideration is crucial for the process. The *Canadian Protected and Conserved Database (CPCAD)* further informs the conservation status of the landscape which ensures that efforts to identify new OECMs complement existing protected areas in order to maintain a level of cohesion and connectivity between protected areas.

Collectively, these datasets are combined to form the foundation of this spatial analysis, allowing for an evidence-based and context-specific approach to identifying high-priority areas for potential OECM recognition in the Átl'ka7tsem/Howe Sound UNESCO Biosphere Region. See Table 1, above, for the list of datasets used in this analysis where the source of the dataset is given along with a brief description.

## 3. Methods

This study employs a structured, spatially explicit approach to identify candidate parcels for Other Effective area-based Conservation Measures within the Átl’ḵa7tsem/Howe Sound Region. The workflow (Figure 2) integrates multiple spatial datasets (Table 1) into different domains, or areas within the workflow. The analysis is designed to prioritize areas that hold high conservation value while also being feasible for OECM designation based on land-use, ownership and zoning attributes.

The following analysis tool place entirely using ArcGIS Pro (version 3.4.2; Esri, 2024), beginning with the process of clipping all the incoming datasets to the defined Átl’ḵa7tsem/Howe Sound Biosphere study area. Ecological and landscape – which include the Critical Habitat for Federally Listed Species at Risk, Species and Ecosystems at Risk, forest age, slope, wetlands, community watersheds, freshwater streams and proximity to existing protected areas – are synthesized into a Weighted Capability Model. Each later was assigned a weight reflecting its perceived importance, its ecological importance and contribution to in-situ biodiversity conservation outcomes (see Table 2) with the total weight summing to 1.0.

The highest weights were given to Critical Habitat for Federally Listed Species at Risk and Species and Ecosystems at Risk (both with a value of 0.2) as they provide a direct link to conservation priorities. Wetlands followed with a weight of 0.15 due to their role in providing additional critical ecosystem services (MacKenzie & Shaw, n.d.). Forest age, watersheds, freshwater streams and proximity to protected areas each contributed equally (0.1), with the slope (areas with a slope greater than 30 degrees) reflecting its more indirect influence on conservation suitability. The forest age, derived from the Vegetation Resource Inventory, is given a normalized, proportional weight based on the age of the forest, and is the only non-binary weighted layer in the suitability model. In other words, the older the forest, the higher the value, with a maximum value of 1, and then multiplied

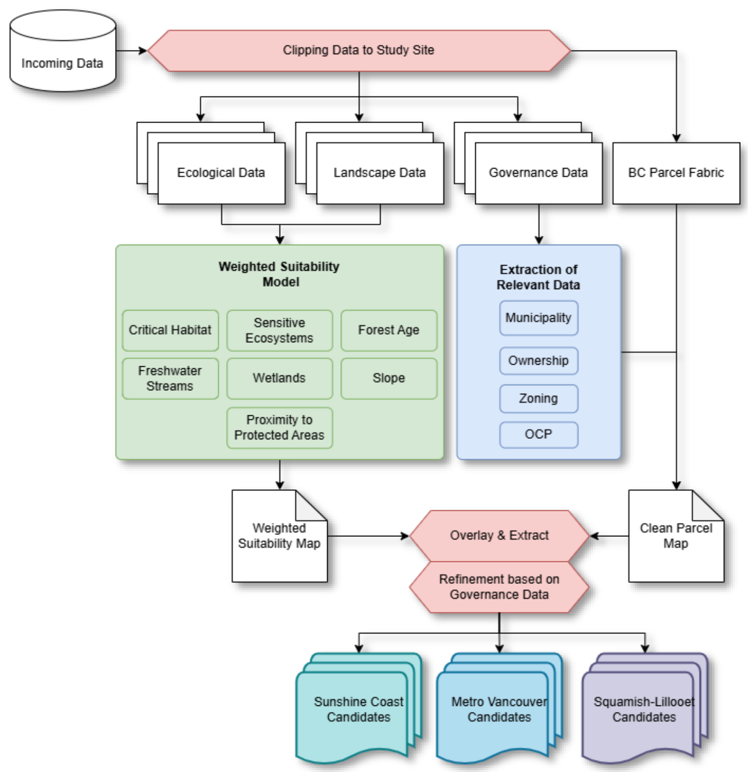


**Figure 2**: The flowchart outlines a workflow for identifying OECM candidates using spatial data. Incoming datasets, including ecological, landscape, governance, and parcel data, are clipped to the study site and processed into a Weighted Capability Model and a Clean Parcel Map. These maps are overlaid and analyzed to identify and sort potential OECM parcels based on ecological value and governance attributes. The final candidates are categorized by administrative regions, supporting conservation planning and decision-making.

by the weight factor. These weighted layers are merged into a hexagonal mesh where each hexagonal polygon acts as a spatial unit for the analysis, with an equal area of 25,980.765 $m^2$. This hexagon size was selected to balance spatial resolution and computational efficiency.

Following the creation of the weighted suitability model, the governance data, which includes the defined parcels, ownership types, zoning classification and parcel classes, are organized in a way that informs the administrative properties of each parcel. This data is essential for the refinement of the initial candidate selection based on the weighted suitability model to ensure the parcels meet the legal requirements of official OECM framework.

The next phase involves overlaying the weighted suitability model with the clean parcel map that has the available zoning data joined spatially. Parcels intersecting the weighted suitability model where the ecological value is high are flagged and initial OECM candidates. These selected parcels are then filtered through a refinement process which uses the governance attributes. The goal is to exclude parcels that are unlikely to meet OECM standards as defined in the Decision Support Tool (DST) which can be found in Mackinnon et al., 2015 and online on government of Canada website. Key considerations in this refinement process include land ownership, parcel class, and zoning laws.

**Table 2. Assigned weights for each input layer used in the weighted suitability model.**

| Layer | Weight[1] |
|---|---|
| Critical Habitat for Federally Listed Species at Risk | 0.2 |
| Species and Ecosystems at Risk | 0.2 |
| Forest Age[2] | 0.1 |
| Wetlands | 0.15 |
| Community Watersheds | 0.1 |
| Fresh Water Streams | 0.1 |
| Park Buffer | 0.1 |
| Slope (> 30º) | 0.05 |
| Total | 1.00 |

[1] Weight values are based on the author's informed judgement and reflect the perceived ecological importance of each variable

[2] Forest weight is a normalized product of the forest age and the assigned weight – older forests will have a higher weight than younger forests.

## 4. Results

The following results are broken up to match the methodology. First presented is the outcome of the suitability model, showing the spatial variation between areas in Átl'k̲a7tsem/Howe Sound that contain more ecological values based on the available data. Following that is the refinement process based on land ownership, class and zoning laws. Finally, the results are broken down based on regional districts to highlight where the most suitable OECM candidates are located.

## *4.1. Weighted Suitability Analysis*

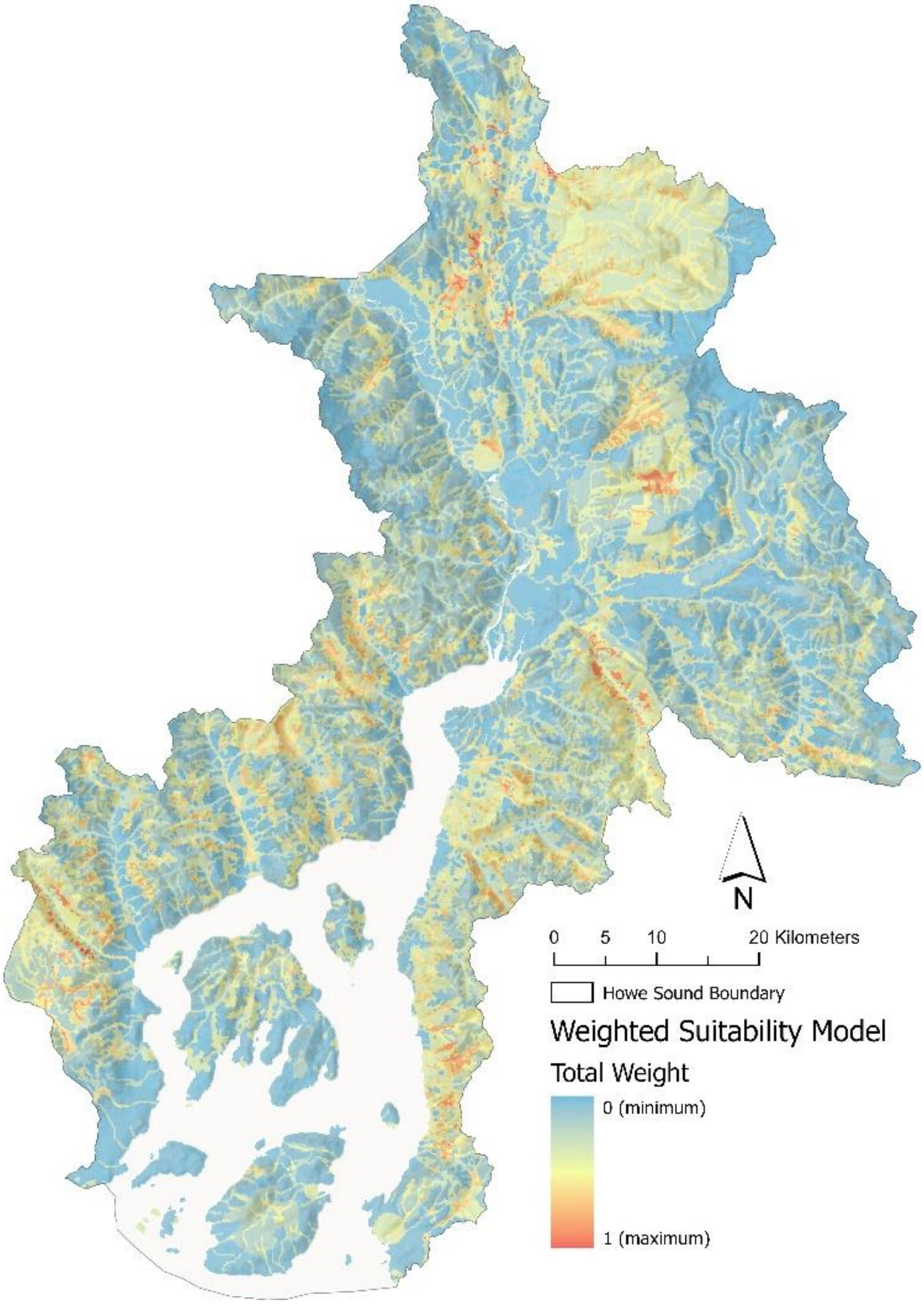


**Figure 3**: Weighted Suitability model for the Átl’ḵa7tsem/Howe Sound Biosphere Region. The map shows the total suitability scores derived from the 8 weighted ecological and landscape datasets. Warmer colors (yellow to red) represent areas with higher cumulative value and cooler colors (blue) indicate locations of lower value. This model is used to locate OECM potential based on the BC Parcel Fabric.

The final suitability surface, shown in figure 3 below, illustrates the spatial distribution of ecolgical value across the biosphere. The scale of the model shows blue hues with lower ecological importance, indicating areas with lower scoring values, and warmer colors in the yellow to red range representing areas of higher combined ecological value. Overall, areas of high value based on the model are well destributed across the biosphere and tend exhibit the highest values when layers overlap, such as locations with wetlands, old forests and the presence of either, or both critical habitat layers.

High values within the model were found in various places, generally away from human settlements and along steep mountainous slopes where little logging forest ages were high, many small streams flow from the mountain tops and where modelled habitat occurred. These areas were found along the Sunshine Coast, west of Port Mellon and Dogpatch, along McNair Creek. There were also some higher values, within Sunshine Coast Regional District, along Dakota Creek. On Bowen Island, some relatively high values are found around Grafton Lake. Notable areas on Gambier Island include Gambier Lake as well as some locations along Gambier Creek. On the mainland, there are some areas of high ecological value based on the model near Cypress Park Falls, along Cypress Creek. Continuing northward, areas of high value are also found along numerous creeks, including Montizambert Creek, Newman Creek, Harvey Creek, and Mineral Creek, near Britannia Beach.

Near Squamish, there are some areas of high value along the Stawamus River, particularly around Ray Creek as well as Northeastern slopes of Sky Pilot Mountain. Continuing northward in the Squamish-Lillooet Regional District, high values were found along Mashiter Creek, and in the far northern part of the regional district, along Rubble Creek and Roe Creek. In all three regional districts in Átl’ḵa7tsem/Howe Sound, areas of high value are scattered around where multiple different layers combine. Two polygons came up with an equal, highest score. The first is located in a wetland area in the Sunshine Coast roughly 13 km NNE from Gibsons, and the second is found near Butterfly Lake in the Squamish-Lillooet regional district, roughly 2.5 kilometers east of the Sea to Sky Highway going towards Whistler.

The parcel selection results are presented based on the three regional districts found in Átl’ḵa7tsem/Howe Sound – Sunshine Coast, Metro Vancouver and Squamish-Lillooet Regional Districts. This is to provide a zoomed in, localized insight into the potential OECM.

## *4.1. Metro Vancouver Regional District*

Within the regional district of Metro Vancouver within Howe Sound Biosphere Region, there were a total of 63 parcels of the current 7,887 parcels selected. This represents less than 1% of all the available parcels, but accounts for over 23% of the parceled area. The major contributor for that land coverage are the subdivision parcels south of Cypress Park, and are mostly under the local government ownership and fall under the CU2 zoning also known as a Community Use Zone (Westvancouver_zoning_bylaws, n.d.). Other parcels of interest include the subdivision parcels also with local government ownership around Killarney Lake on Bowen Island. These generally fall under the zoning P1, a Community Nature Park zoning bylaw where the permittance of new construction is limited (Gambier_zoning_bylaws, n.d.) and discussed later. The last notable inclusion is Lighthouse Park, a parcel already classified as "Park" land and is not currently recognized under federal protection frameworks.

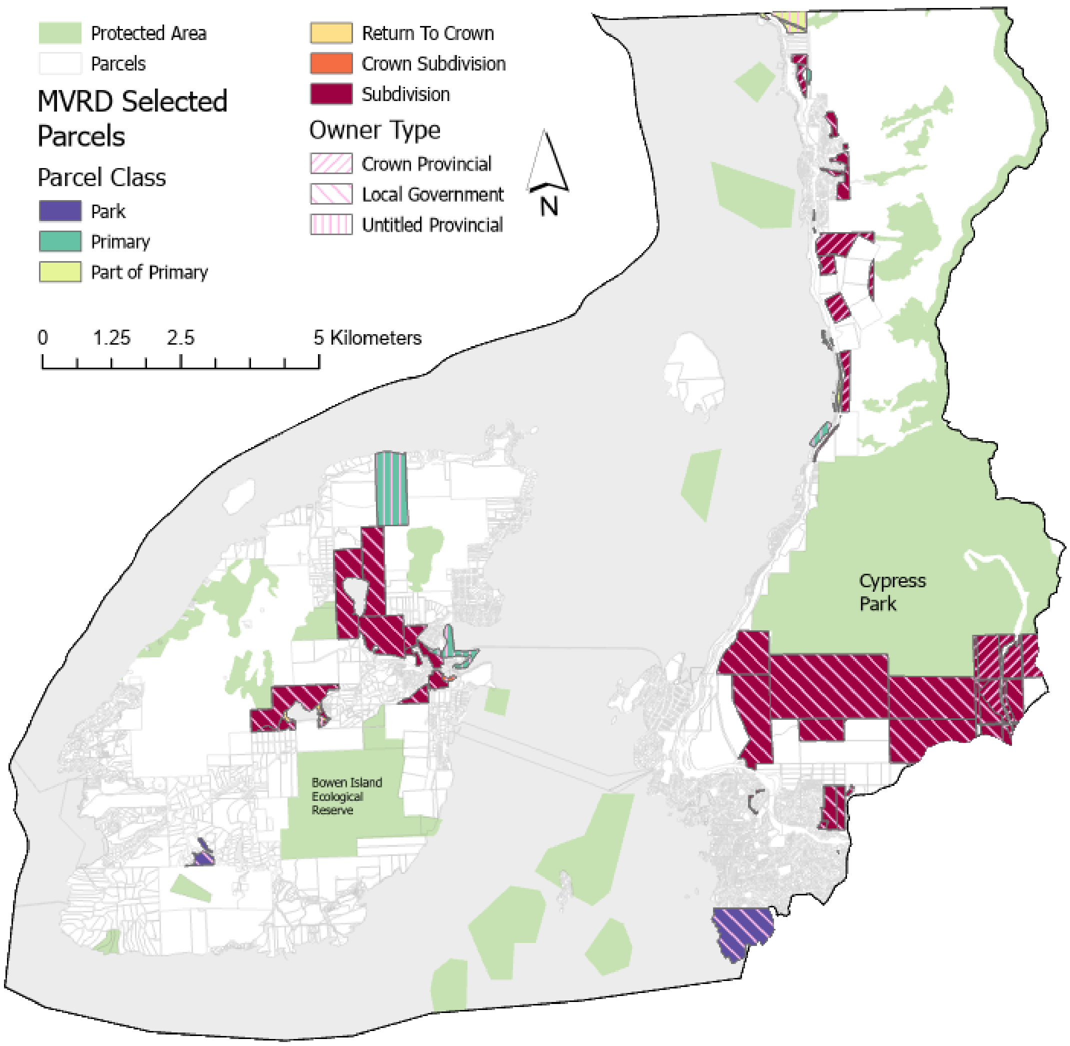


**Figure 4**: Selected OECM candidate parcels within Metro Vancouver Regional District symbolized by parcel class and owner type. The currently protected areas, in green, are also included for context.

## 4.2. *Sunshine Coast Regional District*

The portion of the Sunshine Coast Regional District within the boundary of Howe Sound Biosphere returned a final selection of 45 parcels. The total area of these parcels equates to just shy of 750 hectares, which represents approximately 2% of the total parceled area. Most of the selected parcels are on Gambier Island, with the majority of those falling under the zoning bylaw G1 (Wilderness Conservation). Along the coast are 17 parcels all of which are zones Rural Forest (RF). See figure 5 for the locations of selected parcels in the Sunshine Coast Regional District.

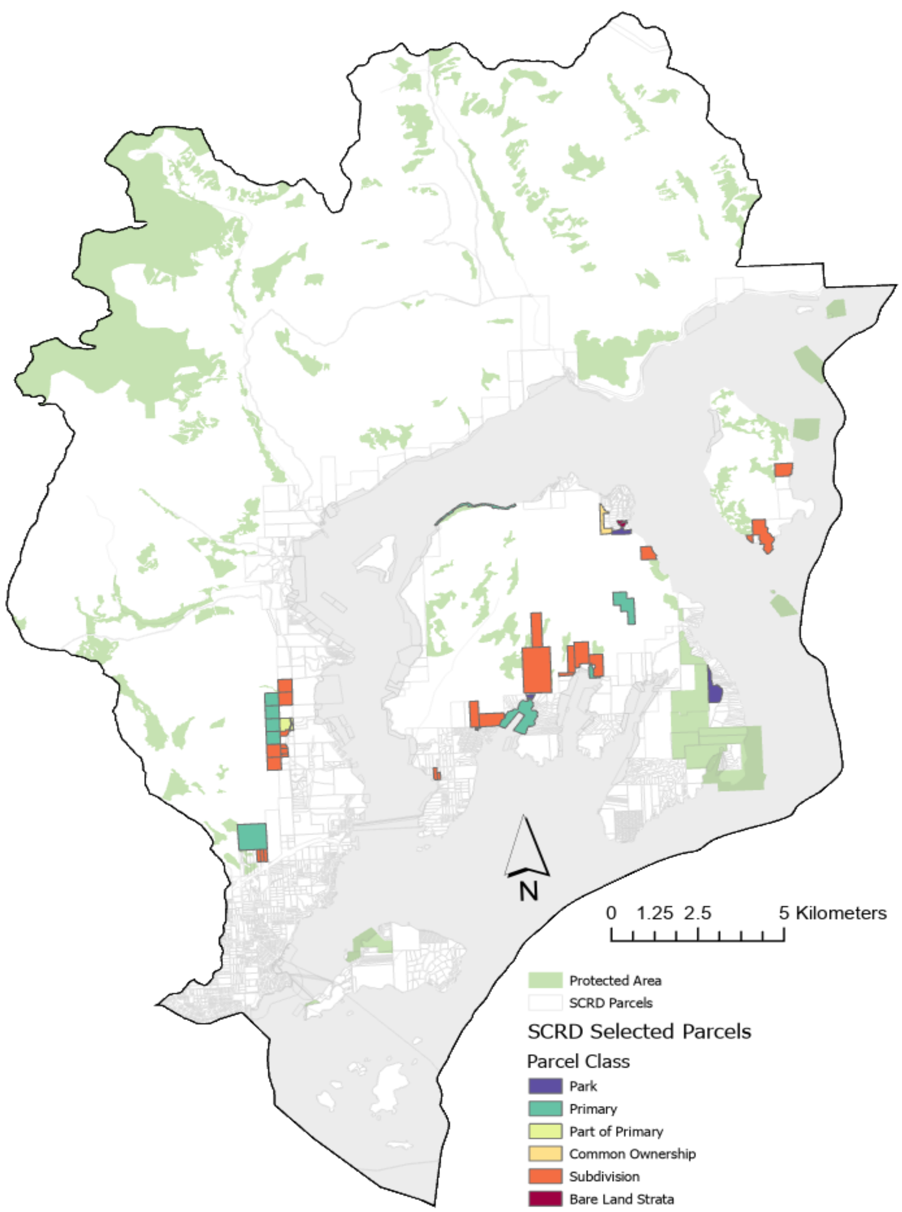


**Figure 5:** Locations of the selected OECM candidate parcels within Sunshine Coast Regional District symbolized by parcel class.

## *4.3. Squamish-Lillooet Regional District*

This third regional district in Howe Sound Biosphere contained the highest amount of selected OECM candidate parcels, likely due to its large area and comparatively more parcels. The total count came to a total of 152 parcels. Of that total the majority are Crown Subdivision and Primary, with 12 parcels selected under Park class. More interestingly are the two local government subdivision parcels across the Squamish River bordering the Skwelwil'Em Squamish Estuary. Another identified parcel of interest boarders the southern portion of Alice Lake, a Primary, Untitled Provincial parcel.

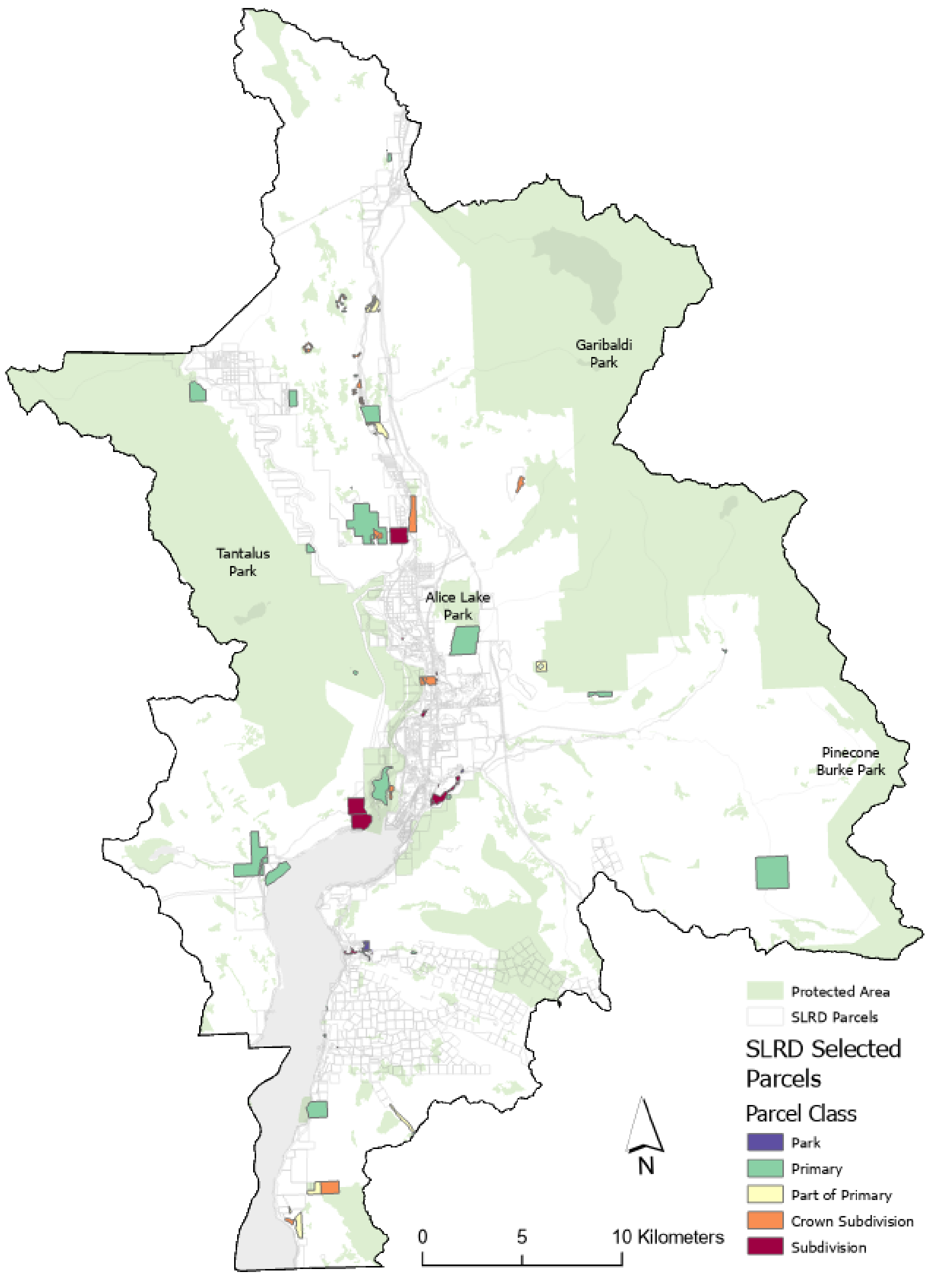


**Figure 6:** Locations of the selected OECM candidate parcels within Squamish-Lillooet Regional District symbolized by parcel class.

**Table 3. Resulting proportion of selected OECM candidate parcels based on the total land and parceled land of each Regional District.**

| (hectares) | SCRD | MVRD | SLRD |
|---|---|---|---|
| Total Land | 56765 | 22151 | 138730 |
| Parceled Land | 36905 | 6156 | 36905 |
| Candidate OECM | 749 | 1455 | 2045 |
| % Of Parceled | **2.0** | **23.6** | **5.5** |

## 5. Discussion

This research aimed to create a methodology used in defining geographic boundaries, based on a given parcel map for Other Effective Area-based Conservation Measures (OECM) using Howe Sound as a case study. By identifying areas where biodiversity conservation can occur as an indirect outcome, a criterion for OECM recognition, the method provides a pragmatic way to contribute to Canada's ambitious conservation goals. This research sought to answer the question: How can geographic information systems (GIS) enhance the process of identifying potential OECMs in Canada? Through the development of a robust and scalable GIS methodology like this, and aligning it with Canada's conservation goals, this research delivers a systematic and objective framework for the selection of candidate OECMs.

The importance of OECM in Canada's conservation strategy is not to be overlooked. Canada has stated its commitment to protect 25% of terrestrial and marine areas by 2025, with an even more ambitious goal of 30% by 2030 (Mansuy et al., 2023). As Canadian lands contain rich diversity, it is necessary to create flexible and innovative conservation strategies (Gurney et al., 2021). OECMs represent an opportunity to designate previously unprotected lands by allowing for areas that are not primarily designated for conservation to contribute to meaningful biodiversity objectives. The results of this study highlight how GIS-based analysis can easily and significantly streamline the effectiveness of identifying new parcels that can contribute to Canada's conservation goals. This study fills in that gap by demonstrating the effectiveness of a structured, data-driven GIS approach. Through the incorporation of available ecological variables such as critical habitat, wetlands, forest age, land tenure and zoning laws into a weighted suitability model, the analysis systematically identifies parcels that not only hold ecological value but also integrates the governing bodies behind each parcel.

The results of this study demonstrate that the methods used effectively meet the core OECM criteria, such as in-situ biodiversity conservation, compatibility with existing land uses, precise boundary delineation, and alignment with a governance structure. This method is also scalable beyond Howe Sound Biosphere, given its controlled and clear approach. Similar spatial analysis and refinement strategies could easily be applied to other regions of Canada and across the world, so long there are ecological datasets, land governance information and proper parcel delineation. The developed framework holds considerable promise as a replicable model that could significantly enhance conservation efforts and efficiency.

## 5.1. *Evaluating the Weighted Suitability Model*

The weighted suitability model developed in this study is the guiding framework and foundational tool for identifying areas where high ecological value and critical habitat features exist. Through the integration of multiple datasets into a mesh of weighted, spatially explicit information, the model provides an objective and scalable framework applicable to a wide range of landscapes, as long as ecological data is present. The synthesis of such a model is not only straightforward, but also quite flexible as additional layers can be incorporated by assigning weights and spatially joining the new layers and their attributes to the existing model. Once included, the model can be easily recalibrated to produce an output between 0 and 1, ensuring consistency across analysis.

One key area for future modification and refinement is the assignment of weights. As mentioned in this study, weights were determined though informed judgement based on previous knowledge of the layer's ecological relevance, but the model would benefit from a more standardized approach to assigning weights. For example, if there was a conservation impact scoring system or literature-derived criteria, it could aid in process of this analysis. Establishing a repeatable method for weight calibration would enhance both the objectivity and replicability of this type of analysis.

The grid size also presents opportunities for experimentation. This study used hexagons with an equal area of 25,980.765 $m^2$ (100-meter edges) in order to try to balance spatial resolution, computational efficiency as well as loading speeds for web mapping applications. Exploring different grid sizes or geometries (diamonds, squares or triangles) could reveal how scale and shape might influence the model sensitivity, which could alter the parcel selection. Ultimately, the chosen grid size and geometry should reflect the context and scale of the analysis.

Overall, the weighted suitability model proved effective in capturing the geometries of the ecological layers as well as its interface with the BC parcel fabric. It was able to successfully locate parcels that have not been previously identified by the Howe Sound Biosphere Region Initiative Society, demonstrating itself as a viable decision-support tool for OECM consideration.

## 5.2. *OECM Candidates Across Regional Districts*

Several important factors must be considered when interpreting the results of this study: the parcel and zoning coverage across the regional districts, as well as each district's capacity and priorities for investing time, funding and administrative resources to OECM recognition. Additionally, it is worth acknowledging the relatively high coverage of Protected Areas and OECMs in Átl'k̲a7tsem/Howe Sound Biosphere Region which may influence future conservation opportunities.

Parcel coverage varies considerably between regional districts. For SCRD, MVRD and SLRD, the proportion of parcel coverage to the overall land area (including marine and freshwater bodies) is approximately 17%, 28% and 15% respectively. Because the analysis relies purely on the BC Parcel Fabric to delineate candidate OECM areas, the extent of parcel coverage significantly contains the available area for consideration. Considering the summary statistics found in Table 3, unsurprisingly MVRD returned a higher proportion of selected parcels that could be considered as OECM as the parcel coverage is higher than the other regional districts. In fact, the proportion of candidate OECMs for each regional district follows the parcel coverage quite closely. Following this, it would be beneficial, from an land conservation approach, that more land is parcelized in order to maximize the potential application of land conservation when one of the requirements for OECM is a ‘clear, defined geographical space with agreed and demarcated borders’ (MacKinnon et al., 2015).

Zoning coverage presents a similar challenge albeit not as severe in this study as it does not affect the parcel selection. OECMs relies heavily on the necessity of having all relevant governing authorities and management mechanisms that align with conservation (J. A. Fitzsimons et al., 2024), so having appropriate zoning characteristic attributed the land that has OECM potential can increase the chance of recognition. However, zoning laws are generally bound within the limits of smaller municipalities and do not extend into the further reaches of terrestrial Átl’ḵa7tsem/Howe Sound Biosphere Region. In this study, SCRD tended to have good zoning coverage, as did MVRD, but SCRD severely lacked coverage, each having a spatial coverage of 80%, 49% and 8% respectively. When evaluated relative to the available parcelized land, the proportions increase to 100% zoning coverage for parcels in SCRD and MVRD but remained lower in SLRD at 55%. Since the analysis used parcels and not parcels with available zoning information, this does not prove to be a direct limitation within this study, but zoning information will be crucial during the formal OECM recognition process to demonstrate compatibility with the Decision Support Tool.

Finally, the priorities and capacities of each regional district play a significant role in the success of OECM recognition, shifting over time. Although this analysis took a top-down approach to locate areas with ecological significance and act as a source of potential OECMs, understanding the priorities of each regional district and aligning one’s approach to recommending OECM recognition should be aligned with the goals of the regional district in which the parcel falls under. A collaborative and context-aware approach that aligns with conservation recommendations with district-level goals is more likely to gain traction and lead to successful OECM recognition.

### *5.3. Considerations*

While the use of GIS undoubtably improves the objectivity and efficiency of OECM candidate selection, it is not without its limitations. Data availability and quality, both spatially and temporally, can pose significant hurdles. Incomplete or outdated datasets can compromise analytical outcomes and affect the validity and robustness of the model. Accurate and current data on land parcels, governing authorities, biodiversity and critical habitats, hydrological networks, vegetation age and health along with whatever other variables that are integrated with the suitability model are critical for valid results. Furthermore, the alignment with land stakeholders and the practicality of implementing OECM remains, as once the candidate has been defined, the process of recognizing it officially is still laborious. GIS can provide land use planners and governments towards areas of higher conservation potential, but the practical implementation depends on the alignment with the local governing structures, land ownership, existing zoning and broader stakeholder reception.

Beyond ecological data improvements, there is much room for improvement. Integrating Indigenous knowledge and community generated data, such as citizen science platforms (eBird or iNaturalist for example) could enrich datasets while ensuring inclusivity and build a stronger support among stakeholders. Alongside with new data integration, future research can also incorporate manual polygon delineation, essentially creating new parcel boundaries within areas of high ecological value, a major limitation of this study, so long there is governing structures and zoning laws within those new parcels. The logical next step would be to expand this framework to other diverse landscapes in Canada in order to test its applicability, scalability and validity.

## 6. Conclusion

In conclusion, this study demonstrates that GIS and spatial analysis can significantly enhance the process of potential OECM identification by using an objective, scalable and replicable framework. The methodology in this study bridges the gap between spatially available data and science, ecological conservation strategies, governing considerations, offering a replicable model for conservation planning. As Canada pursues its ambitious conservation targets, incorporating GIS approaches like in this study will be crucial to achieving the nation’s biodiversity and climate objectives.

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